

Exercises for *Secular Coefficients* mini-course

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June 1–6, 2026

1 Background

1.1 Number theory

1. Prove that $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$ holds for $\Re s > 1$.
Given a nontrivial Dirichlet character χ modulo m , prove that $L(s, \chi) = \prod_p (1 - \chi(p)p^{-s})^{-1}$ for $\Re s > 1$.
2. Show that $\zeta(s) - 1/(s-1)$ has analytic continuation to $\Re s > 0$.
Hint: express the difference as $1 - s \int_1^\infty \{t\} t^{-s-1} dt$ when $\Re s > 1$, where $\{t\} := t - \lfloor t \rfloor$ is the fractional part of t .
3. Given a nontrivial Dirichlet character χ modulo m , prove that $\sum_{a \in (\mathbb{Z}/m\mathbb{Z})^\times} \chi(a) = 0$.
Prove that $\sum_{\chi \bmod m} \chi(a) \overline{\chi}(b) = \phi(m) \mathbf{1}_{a \equiv b \bmod m}$ where the sum is over all Dirichlet characters modulo m (here $(a, m) = (b, m) = 1$).
4. Given a nontrivial Dirichlet character χ modulo m , prove that $\sum_{n=1}^\infty \chi(n) n^{-s}$ converges (conditionally) for $\Re s > 0$.
Hint: observe there is a constant C such that $|\sum_{n \leq x} \chi(n)| \leq C$ holds for all $x \geq 1$.
5. Let M be a polynomial in $\mathbb{F}_q[T] \setminus \mathbb{F}_q$. Given a nontrivial Dirichlet character χ modulo M , prove that $\sum_{A \in (\mathbb{F}_q[T]/M\mathbb{F}_q[T])^\times} \chi(A) = 0$.
Deduce that $\mathcal{L}(u, \chi) = \sum_{f \text{ monic}} \chi(f) u^{\deg f}$ is a polynomial of degree at most $\deg M - 1$.
Show that if $\mathcal{L}(u, \chi) = \prod_{i=1}^d (1 - \gamma_i u)$ for complex numbers $\gamma_1, \dots, \gamma_d$ then

$$\sum_{f \in \mathbb{F}_q[T], f \text{ monic}, \deg f = n} \chi(f) \Lambda(f) = - \sum_{i=1}^d \gamma_i^n$$

where $\Lambda(f) = \deg P$ if f is a power of an irreducible polynomial P , and 0 otherwise.

1.2 Probability

1. If $X \sim N(0, 1)$ (X is distributed like a standard real Gaussian) then, for every real a ,
$$\mathbb{E}[e^{aX}] = e^{a^2/2}.$$
2. If $X \sim N_{\mathbb{C}}(0, 1)$ (X is distributed like a standard complex Gaussian) then, for every real a, b ,
$$\mathbb{E}[e^{aX + b\overline{X}}] = e^{ab}.$$
3. If $X \sim N_{\mathbb{C}}(0, 1)$ then $\mathbb{E}[X^{k_1} \overline{X}^{k_2}] = (k_1)! \delta_{k_1, k_2}$, and $\mathbb{E}[X^{k_1} \overline{X}^{k_2}] = 0$ otherwise.

2 Tutorial exercises

1. Prove that $\mathbb{E}[\alpha(n)\overline{\alpha}(m)] = 0$ if n, m are distinct positive integers, and $\mathbb{E}[\alpha(n)\overline{\alpha}(m)] = 1$ if $n = m$. Here α is the Steinhaus random multiplicative function.
2. Prove that $\mathbb{E} \left| \frac{1}{\sqrt{x}} \sum_{n \leq x} \alpha(n) \right|^4 \sim C \log x$ for an explicit constant $C > 0$.
Hint: parametrize solutions to $ab = cd$ by $a = ef, b = gh, c = eg, d = fh$ with $\gcd(f, g) = 1$.
3. Let $U_N \in U(N)$ be a random Haar-distributed unitary matrix. Define $\text{Sc}_i(U_N)$ via $\det(I - uU_N) = \sum_{i=0}^N (-u)^i \text{Sc}_i(U_N)$. For each $0 \leq n \leq N$ show that $|\text{Sc}_n(U_N)|$ and $|\text{Sc}_{N-n}(U_N)|$ have the same law.
Hint: Use the functional equation.
4. Let $(N_i)_{i \geq 1}$ be iid standard complex Gaussians. Let $F(u) = \exp(-\sum_{i \geq 1} N_i u^i / \sqrt{i})$. Denote by c_n the coefficient of u^n in F . Prove that $\mathbb{E}|c_n|^4 = n + 1$.
Hint: Consider $\mathbb{E}[F(u_1)F(u_2)\overline{F}(u_3)\overline{F}(u_4)]$. Can you compute higher moments? What do you get?
5. Compute $\mathbb{E}|L(1/2, \chi_M)|^2$ asymptotically (as $\deg M \rightarrow \infty$), where χ_M is a Dirichlet character chosen uniformly at random from the set of nontrivial Dirichlet characters modulo an irreducible $M \in \mathbb{F}_q[T]$. Can you compute $\mathbb{E}|\det(I - U_N)|^2$ where $U_N \in U(N)$ is a Haar-distributed unitary matrix?
6. Let χ_M be as in the last exercise. Recall that $\mathcal{L}(u, \chi_M) = \det(I - u\sqrt{q}\Theta_{\chi_M})$ for some matrix $\Theta_{\chi_M} \in U(\deg M - 1)$. Fix $k \geq 1$. Show that if $nk < \deg M$ then $\mathbb{E} \left| \frac{\text{Tr}(\Theta_{\chi_M}^n)}{\sqrt{n}} \right|^{2k} \sim \mathbb{E}|N_{\mathbb{C}}(0, 1)|^{2k}$ as $\deg M \rightarrow \infty$.

3 More

1. For x, y on the unit circle, estimate $\mathbb{E}[G_N(x)G_N(y)]$ where $G_N(x) = \Re \sum_{i=1}^N \frac{N_i}{\sqrt{i}} x^i$, and $(N_i)_i$ are iid standard complex Gaussians. For x, y with real part equal to $1/2$ estimate $\mathbb{E}[\Re(\log F_N(x))(\Re(\log F_N(y)))]$ where $F_N(s) = \prod_{p \leq N} (1 - \alpha(p)p^{-s})^{-1}$, α the Steinhaus function.
2. (Bohr–Jessen in $\Re s > 1$.) Let α be the Steinhaus function and let $F(s) = \sum_{n \geq 1} \alpha(n)/n^s$. Prove that $\zeta(s + iU_T) \xrightarrow{d} F(s)$ as $T \rightarrow \infty$, where s is fixed with $\Re s > 1$, and $U_T \sim \text{Uniform}[0, T]$.
3. (Bohr–Jessen in $\Re s > 1$ for Dirichlet characters over a finite field.) Let χ_M be a Dirichlet character chosen uniformly at random from the set of nontrivial Dirichlet characters modulo an irreducible $M \in \mathbb{F}_q[T]$. As $\deg M \rightarrow \infty$, show that $L(u, \chi_M) \xrightarrow{d} L(u, \alpha_q)$ for every $|u| < 1/q$, where α_q is the Steinhaus random multiplicative function defined on monic polynomials.
4. The following is a precise version of McLeish CLT, extracted from the paper of Soundararajan–Xu.

Theorem (McLeish). *Let $Z_1, \dots, Z_N$ be a martingale difference sequence of complex-valued random variables, i.e. $S_N = \sum_{n=1}^N Z_n$ is a martingale. For any real $t_1, t_2 \in \mathbb{R}$ we have, with $t^2 = (t_1^2 + t_2^2)/2$,*

$$\begin{aligned} \mathbb{E}[e^{it_1 \text{Re}(S_N) + it_2 \text{Im}(S_N)}] &= e^{-t^2/2} + O\left(e^{t^2} \left(\sum_{n=1}^N \mathbb{E}[|Z_n|^4] \right)^{\frac{1}{4}}\right) + O\left(e^{t^2} \left(\mathbb{E}\left[\left(\sum_{n=1}^N |Z_n|^2 - 1 \right)^2 \right] \right)^{\frac{1}{2}}\right) \\ &\quad + O\left(e^{t^2} \max_{\phi} \left(\mathbb{E}\left[\left(\sum_{n=1}^N (e^{-i\phi} Z_n^2 + e^{i\phi} \overline{Z_n}^2) \right)^2 \right] \right)^{\frac{1}{2}}\right). \end{aligned}$$

Deduce the criterion of Soundararajan–Xu: let $\mathcal{A}_x \subseteq \{1, 2, \dots, x\}$ and set $S(x, \mathcal{A}_x) = \sum_{n \in \mathcal{A}_x} \alpha(n)/\sqrt{|\mathcal{A}_x|}$. Then $S(x, \mathcal{A}_x)$ tends to a standard complex Gaussian if two conditions are met: $\lim_{x \rightarrow \infty} \mathbb{E}|S(x, \mathcal{A}_x)|^4 = 2$, and $|\{n \in \mathcal{A}_x : P(n) = p\}| = o(|\mathcal{A}_x|)$ for every $p \leq x$ ($P(n)$ is the largest prime factor of n).

5. Let $b(\lambda)$ be a function on partitions. Given a partition λ , we denote by $a_i(\lambda)$ the number of parts of λ of size i . Let $S_N = \sum_{\lambda \vdash N} b(\lambda) \prod_{i=1}^N N_i^{a_i(\lambda)}$. Using McLeish CLT, establish a sufficient criterion for S_N to tend in distribution to a standard complex Gaussian.